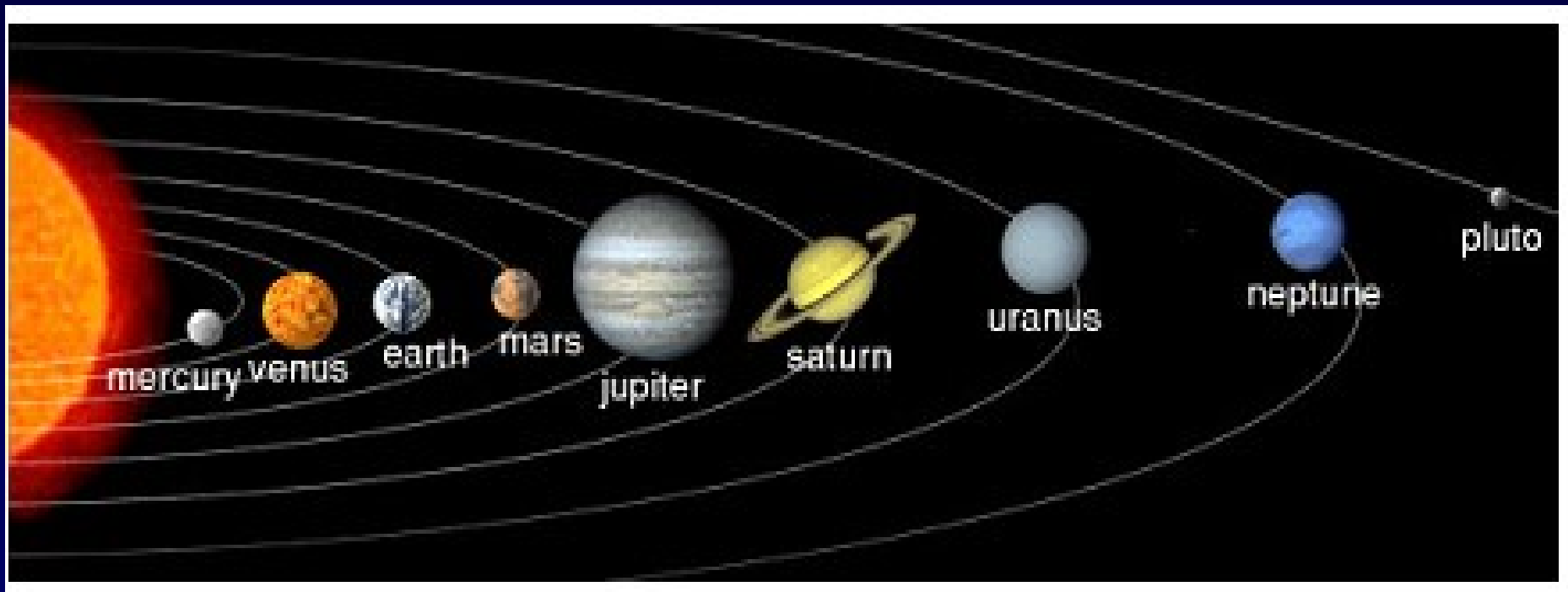


The Laws of Planetary Motion

Kepler's and Bode's Law



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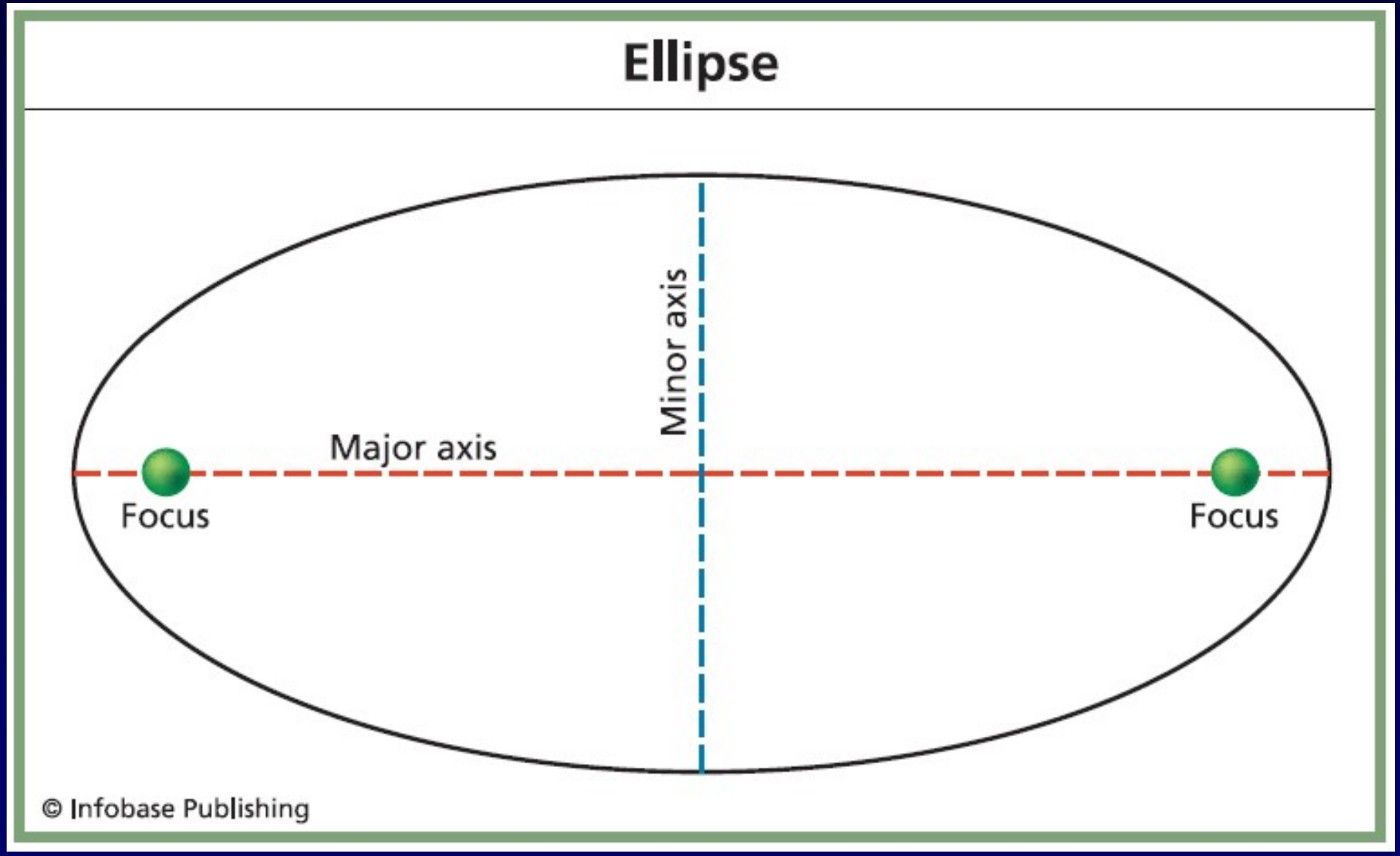
Journey from Geocentric to Heliocentric

- From the ancient Egyptians to the time of Newton, the planets that were first seen as gods ended up as objects in the universe.
- The Earth was initially considered from being the center of the universe (Geocentric) but later considered being just another planet.
- Nicolaus Copernicus had shown that his model – Heliocentric – of the solar system (with all of the planets, including the Earth, orbiting the Sun) did a much better job of explaining the motions of the planets.
- In fact, the Copernican system was gaining more and more acceptance among the scientists of the day. At the same time, however, the church rejected any plan of the universe that did not have the Earth at the center.
- Tycho Brahe, a Danish astronomer, known for his comprehensive astronomical observations, was trying to reconcile these two opposing views - he wanted them both to be right.
- Tycho's scientific reasoning told him that Copernicus made sense. But Tycho also noticed that gravity pulled everything toward the Earth, and he assumed that the Earth pulled on the Sun, the Moon, the other planets, and the stars as well. If everything in the universe (as Tycho saw it) was being pulled toward the Earth, then the Earth simply had to be at the center of everything.

Journey from Geocentric to Heliocentric

- Tycho wanted to find something that would satisfy both his intellect and his religious beliefs. Ultimately, his system did not stand the test of time, and it lasted only a little while.
- But what did remain from Tycho's work were some of the most amazingly precise measurements of planetary positions and motions that had been compiled up to that time.
- These measurements were vitally important to Tycho's student, Johannes Kepler.
- Kepler was, without a doubt, Tycho's greatest student. Tycho collected the data, but Kepler was the one to start to figure out what it all meant.
- It began when Tycho set Kepler to work on the orbit of Mars. Mars was one of the closest planets, and it should have been easy to figure out its orbit. Yet Mars could never be found at its predicted position, and Kepler wanted to know why. After a great deal of work, he finally figured it out and came up with what is known as Kepler's first law of planetary motion at that time.

Johannes Kepler discovered that planetary orbits are ellipses, not circles as once believed.



Johannes Kepler discovery provided answer to his teacher Tycho

- What Kepler found was that planet's orbits are not perfect circles.
- Rather, they are ellipses.
- This was the reason Mars was not in its expected location, as Tycho was expecting it to follow a circle.

Kepler's Laws of Planetary Motion

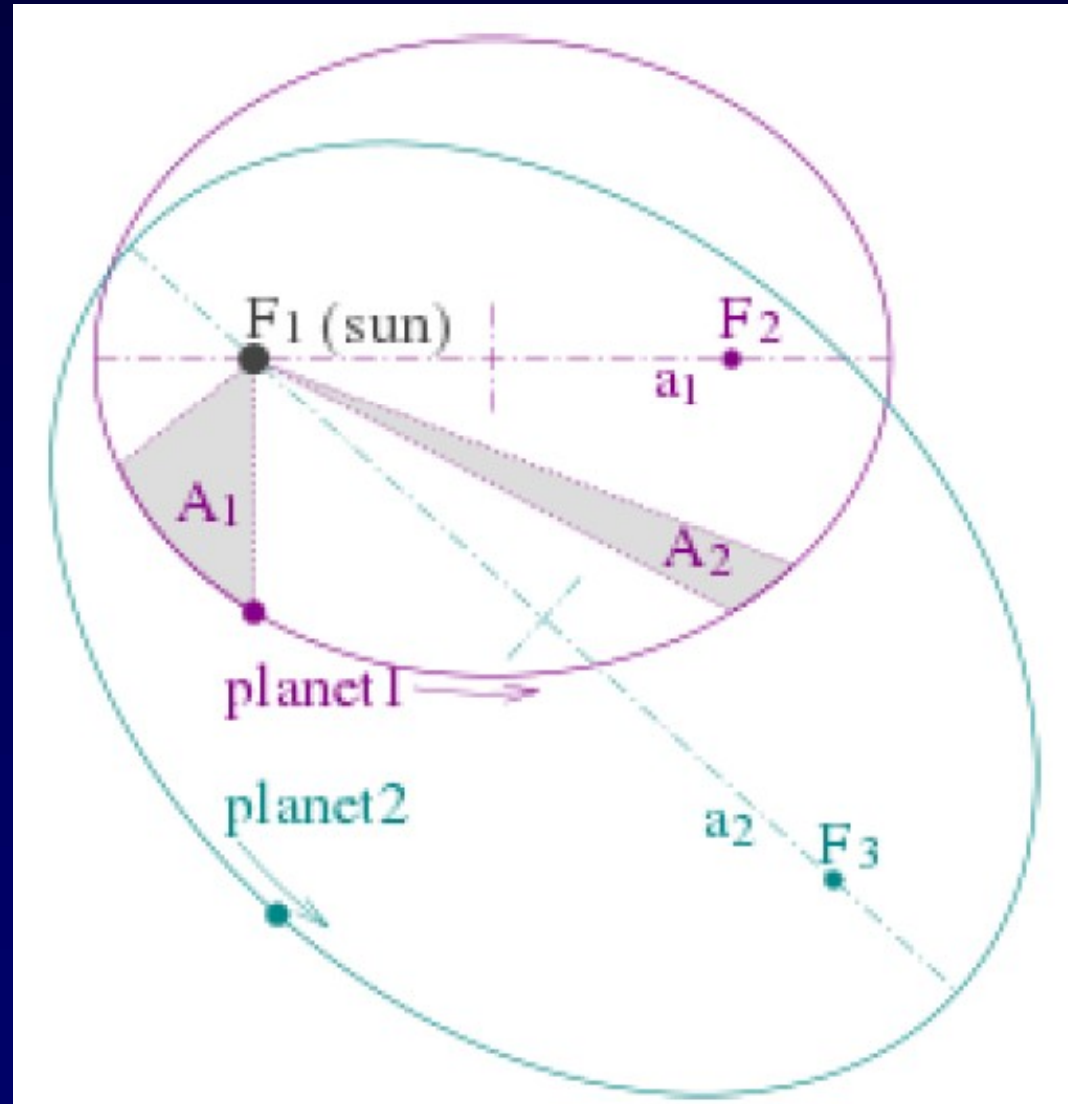
- Kepler's laws of planetary motion was published by Johannes Kepler between 1609 and 1619. The first and second laws were published in 1609 and the third law appeared in 1619.
- It describe the orbits of planets around the Sun.
- The laws modified the heliocentric theory of Nicolaus Copernicus, replacing its circular orbits and epicycles with elliptical trajectories.
- The three laws explains how planetary velocities vary.

Kepler's Laws

The three laws of Kepler's state that:

1. The orbit of a planet is an ellipse with the Sun at one of the two foci.
2. A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.
3. The ratio of the square of a planet's orbital period (T^2) to the cube of the semi-major axis of its orbit (a^3) is constant for all the planets, including the Earth. $\rightarrow T_1^2 / T_2^2 = r_1^3 / r_2^3$

Illustration of Kepler's three laws with two planetary orbits.



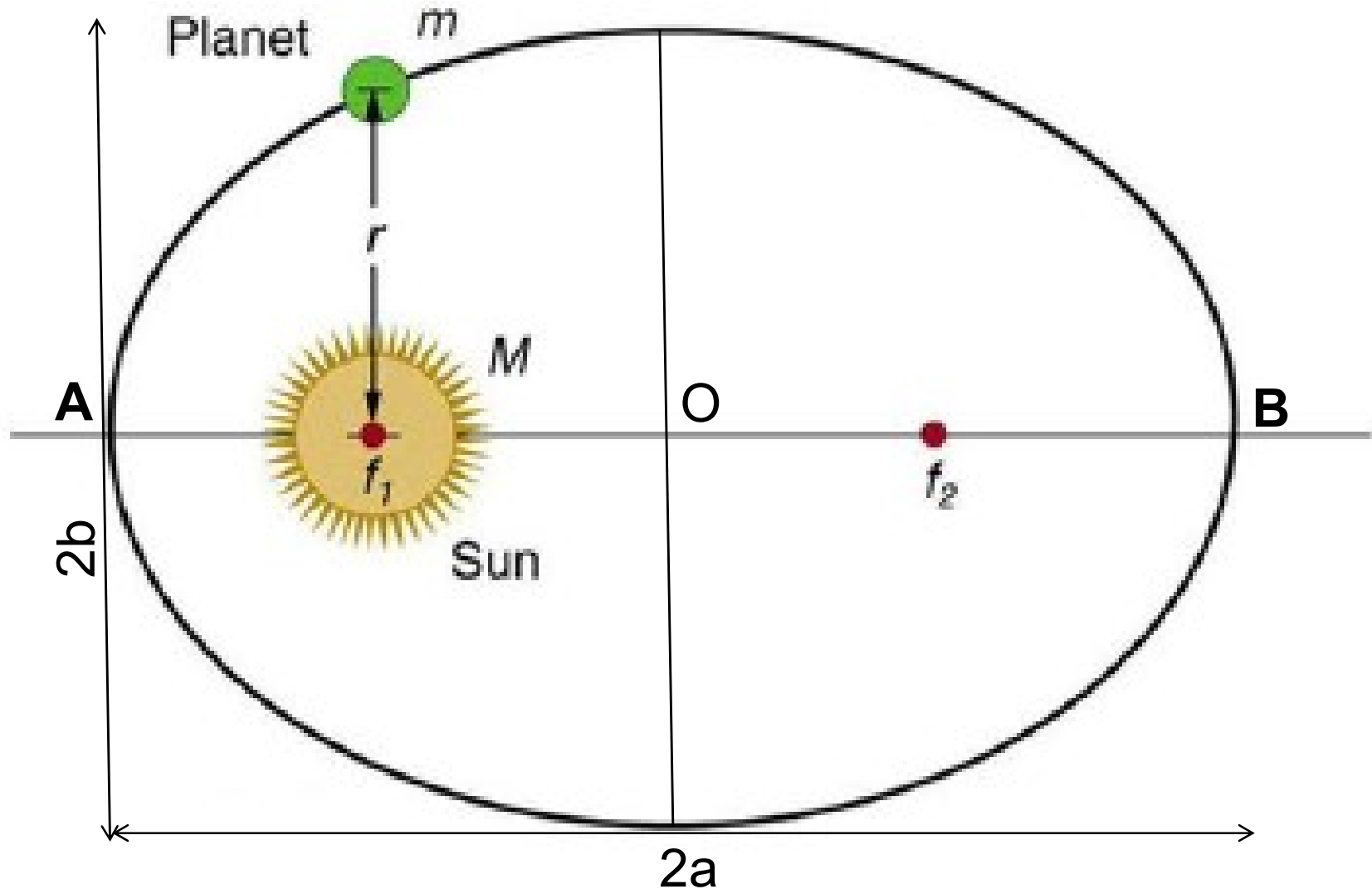
1. The orbits are ellipses, with focal points F1 and F2 for the first planet and F1 and F3 for the second planet. The Sun is placed at focal point F1.
2. The two shaded sectors A1 and A2 have the same surface area and the time for planet 1 to cover segment A1 is equal to the time to cover segment A2.
3. The total orbit times for planet 1 and planet 2 have a ratio of $(a_1/a_2)^{3/2}$.

Johannes Kepler's laws improved the model of Copernicus. If the eccentricities of the planetary orbits are taken as zero, then Kepler basically agreed with Copernicus:

1. The planetary orbit is a circle with epicycles.
2. The Sun is approximately at the center of the orbit.
3. The speed of the planet in the main orbit is constant.

The eccentricities of the orbits of those planets known to Copernicus and Kepler are small, so the foregoing rules give fair approximations of planetary motion, but Kepler's laws fit the observations better than does the model proposed by Copernicus. Kepler's corrections are:

1. The planetary orbit is not a circle with epicycles, but an ellipse.
2. The Sun is not near the center but at a focal point of the elliptical orbit.
3. Neither the linear speed nor the angular speed of the planet in the orbit is constant, but the area speed (closely linked historically with the concept of angular momentum) is constant.



$Of_1 = Of_2 = ae$, where e is the eccentricity.

Therefore distance between points A and f_1 is $a - ae = a(1 - e)$

Similarly distance between points B and f_1 is $a + ae = a(1 + e)$

Perihelion and Aphelion of the Earth

- Perihelion is the point in the orbital pathway of earth where it is closest to the sun.
- The closest distance (D_c) between Sun and the Earth is given by distance between points A and f_1 :
$$D_c = a - ae = a(1 - e)$$
 where eccentricity (e) is given by $e = \sqrt{(a^2 - b^2)}/a$, a and b are semi major and minor axis of ellipse.
- Aphelion on the other hand is the point in the orbit when the earth is farthest away from the sun.
- The farthest distance (D_f) between Sun and the Earth is given by distance between points B and f_1 :
$$D_f = a + ae = a(1 + e)$$

Derivation of Kepler's Third Law

- It is possible to derive Kepler's third law by starting with Newton's laws of motion and the universal law of gravitation.
- It can be therefore demonstrated that the force of gravity is the cause of Kepler's laws.
- Consider a circular orbit of a small mass m around a large mass M . Gravity supplies the centripetal force to mass m . Starting with Newton's second law applied to circular motion,

$$F_{\text{net}} = ma_c = mv^2 / r$$

The net external force on mass m is gravity, and therefore substituting the force of gravity for F_{net} :

$$GmM / r^2 = mv^2 / r \quad \rightarrow \quad GM / r = v^2$$

Derivation of Kepler's Third Law

- Now, to get at Kepler's third law, we must get the period P into the equation. By definition, period T is the time for one complete orbit. Now the average speed v is the circumference divided by the period - that is,

$$v = 2\pi r / T$$

Substituting this into the previous equation gives:

$$GM / r = (2\pi r)^2 / T^2$$

$$T^2 = 4\pi^2 r^3 / GM$$

Using subscripts 1 and 2 to denote two different satellites, and taking the ratio of the last equation for satellite 1 to satellite 2 yields:

$$T_1^2 / T_2^2 = r_1^3 / r_2^3$$

This is Kepler's third law.

Importance of Kepler's Third Law

Kepler's laws were developed for the solar system but are applicable to any closed planetary system. They govern the motion of any natural or artificial satellite about a parent body. Kepler's third law relates the period (T) *and* the semi-major axis (a) *of the orbit of the satellite to the* mass (M) *of the parent body through the equation:*

$$GM = \frac{4\pi^2}{T^2}a^3$$

where G is the gravitational constant. This relationship was extremely important for determining the masses of those planets that have natural satellites. It can now be applied to determine the masses of planets using the orbits of artificial satellites.

Bode's law

- In 1772 the German astronomer Johann Bode devised an empirical formula to express the approximate distances of the planets from the Sun. Bode's law is also known as Titius-Bode law, because the same relationship had been suggested earlier by J. D. Titius of Wittenberg.
- A series of numbers is created in the following way: the first number is zero, the second is 0.3, and the rest are obtained by doubling the previous number. This gives the sequence 0, 0.3, 0.6, 1.2, 2.4, 4.8, 9.6, 19.2, 38.4, 76.8, etc. Each number is then augmented by 0.4 to give the sequence: 0.4, 0.7, 1.0, 1.6, 2.8, 5.2, 10.0, 19.6, 38.8, 77.2, etc. This series can be expressed mathematically as follows:

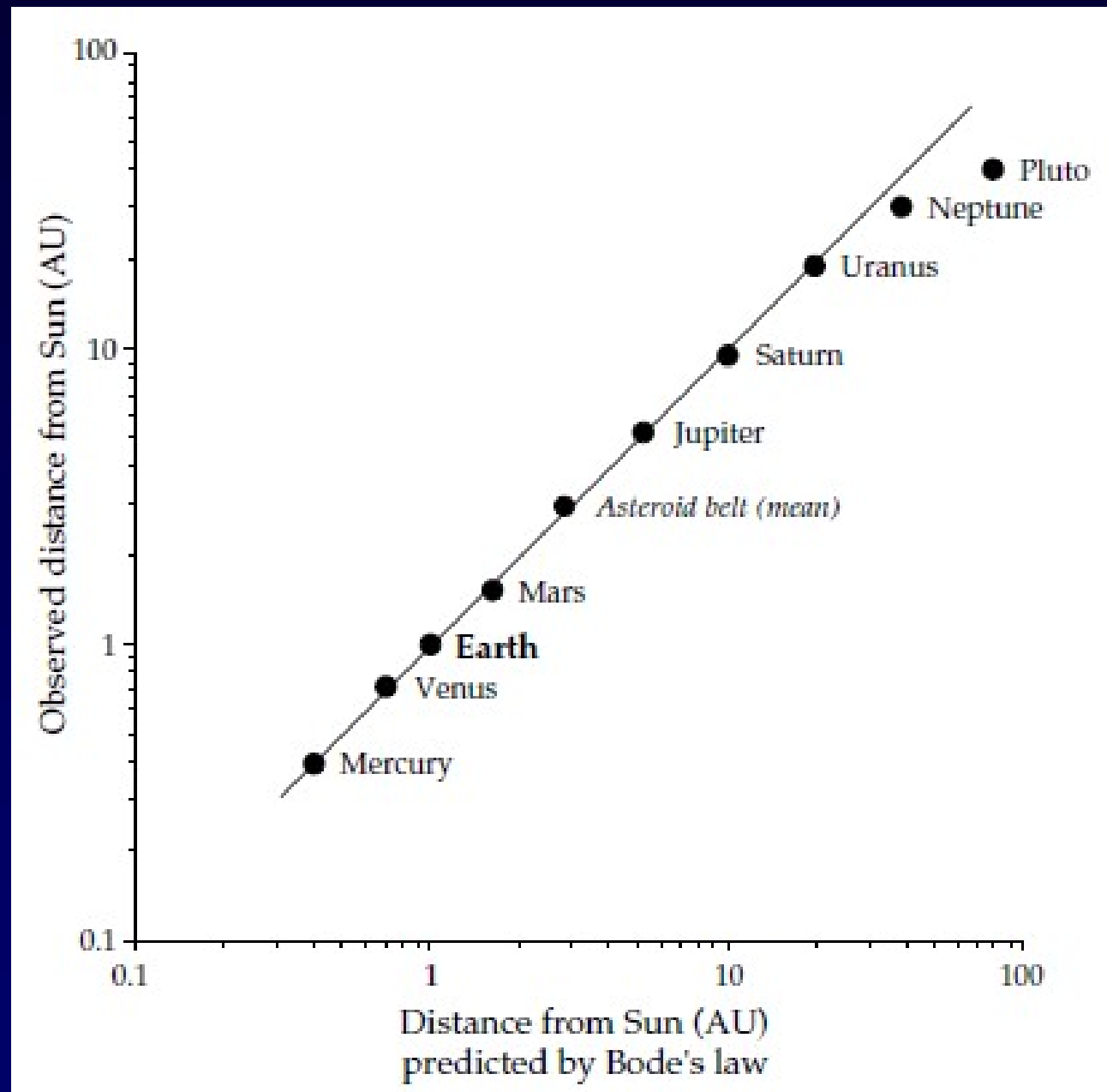
$$d_n = 0.4 \text{ for } n = 1$$

$$d_n = 0.4 + 0.3 \times 2^{(n-2)} \text{ for } n \text{ is more than and equal to } 2.$$

Distance of the planets from the Sun

Planets	Distance in A.U. (Computed)	Distance in A.U. (As per Bode's Law)
1. Mercury	0.39	0.40
2. Venus	0.72	0.70
3. Earth	1.00	1.00
4. Mars	1.52	1.60
5. Asteroid Belt	2.77	2.80
6. Jupiter	5.20	5.20
7. Saturn	9.54	10.00
8. Uranus	19.19	19.60
9. Neptune	30.07	38.8
10. Pluto	39.52	77.2

Bode's empirical law for the distances of the planets from the Sun



Discrepancies in the Bode's Law

It may be noted from the Table showing distances of planets from the Sun and also graph shown in the last slide that the Titius-Bode law holds remarkably well, except for Neptune and Pluto.

A possible interpretation of the discrepancies is that the orbits of these planets are no longer their original orbits.

As we know, Bode's Law for more than seven planets which are away from the Sun can be expressed mathematically as follows:

$$d_n = 0.4 \text{ for } n = 1$$

$$d_n = 0.4 + 0.3 \times 2^{(n-2)} \quad \text{For } n \text{ more than or equal to } 2.$$

Bode's Law for $n = 9$ and 10 , can be expressed mathematically as:

$$d_n = 0.4 + 0.3 \times 2^{(n-2)} - \{(n-8)(3)\}^2 \quad \text{For } n \text{ is more than or equal to } 8.$$

A Fifth Planet or Asteroid Belt → Prediction from the Bode's law

- Bode's law predicts a fifth planet at 2.8 AU from the Sun, between the orbits of Mars and Jupiter.
- In the last years of the eighteenth century astronomers searched intensively for this missing planet. In 1801 a small planetoid, Ceres, was found at a distance of 2.77AU from the Sun.
- Subsequently, it was found that numerous small planetoids occupied a broad band of solar orbits centered about 2.9 AU, now called the *asteroid belt*.
- Pallas was found in 1802, Juno in 1804, and Vesta, the only asteroid that can be seen with the naked eye, was found in 1807. By 1890 more than 300 asteroids had been identified. In 1891 astronomers began to record their paths on photographic plates. Thousands of asteroids occupying a broad belt between Mars and Jupiter, at distances of 2.15 - 3.3 AU from the Sun, have since been tracked and cataloged.



Thank You

Pillow lava at CIR observed during manned submersible dive